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原始脚本的访问地址

https://www.bilibili.com/video/BV1df4y177kB/?spm_id_from=333.788.recommend_more_video.1&vd_source=7573895a5d8ffac9ca22fd9348ac3b22

本文在原始文件的基础上，添加了批量处理原始文件的脚本，并将 GUI 文件名改为通用文件。

文件介绍

共有五个文件，分别为

1、工艺库原始数据文件夹

2、处理好的数据文件（.mat）

3、批量处理工艺原始数据并导入到数据文件的脚本（GmId_readDates.m）

4、GUI 文件（.fig）

5、GUI 文件生成的 m 文件（MOS_GmId_GUI.m）

csmc018_mos18	20 items	folder	Fri 30 Dec 2022 10:03:52 AM CST
GmId_readDates.m	2.6 KB	Objective-C source code	Fri 30 Dec 2022 11:41:26 AM CST
MOS_GmId_Data.mat	282.2 KB	unknown	Fri 30 Dec 2022 11:40:24 AM CST
MOS_GmId_GUI.fig	67.9 KB	XFig image	Fri 30 Dec 2022 10:42:35 AM CST
MOS_GmId_GUI.m	45.0 KB	MATLAB script/function	Fri 30 Dec 2022 10:42:35 AM CST

使用方法

1、对工艺库 MOS 管扫描得到工艺库原始数据，保存为.csv 类型。

（数据扫描过程见下文）

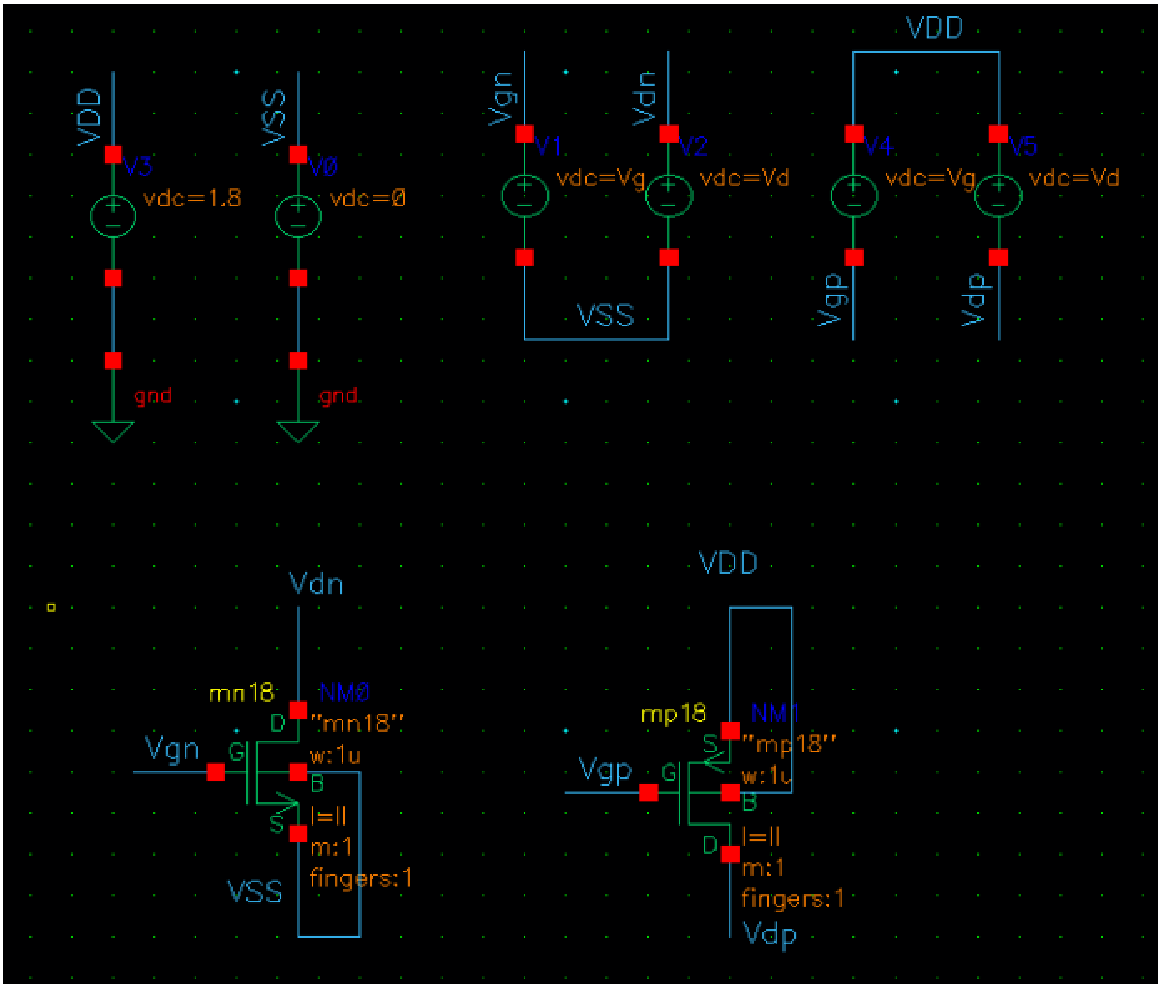
nch18_Cdd_Cgg.csv	81.3 KB	CSV document	Thu 29 Dec 2022 09:41:11 AM CST
nch18_Cgd_Cgg.csv	83.5 KB	CSV document	Thu 29 Dec 2022 01:47:46 PM CST
nch18_ft.csv	79.4 KB	CSV document	Thu 29 Dec 2022 01:50:53 PM CST
nch18_gm_gds.csv	77.3 KB	CSV document	Thu 29 Dec 2022 02:31:41 PM CST
nch18_gm_Id.csv	77.3 KB	CSV document	Thu 29 Dec 2022 10:46:04 AM CST
nch18_Id_W.csv	79.6 KB	CSV document	Thu 29 Dec 2022 01:52:12 PM CST
nch18_Vdsat.csv	80.3 KB	CSV document	Thu 29 Dec 2022 01:57:29 PM CST
nch18_Vgs_Vth.csv	80.6 KB	CSV document	Thu 29 Dec 2022 01:59:54 PM CST
pch18_Cdd_Cgg.csv	81.1 KB	CSV document	Thu 29 Dec 2022 03:05:22 PM CST
pch18_Cgd_Cgg.csv	83.4 KB	CSV document	Thu 29 Dec 2022 03:06:14 PM CST
pch18_ft.csv	79.4 KB	CSV document	Thu 29 Dec 2022 03:07:38 PM CST
pch18_gm_gds.csv	77.3 KB	CSV document	Thu 29 Dec 2022 03:07:11 PM CST
pch18_gm_Id.csv	77.2 KB	CSV document	Thu 29 Dec 2022 03:09:36 PM CST
pch18_Id_W.csv	82.2 KB	CSV document	Thu 29 Dec 2022 03:08:37 PM CST
pch18_Vdsat.csv	82.5 KB	CSV document	Thu 29 Dec 2022 03:09:01 PM CST
pch18_Vgs_Vth.csv	81.3 KB	CSV document	Thu 29 Dec 2022 03:09:58 PM CST
Vds_nch18_gm_gds.csv	135.1 KB	CSV document	Thu 29 Dec 2022 04:25:26 PM CST
Vds_nch18_gm_Id.csv	134.4 KB	CSV document	Thu 29 Dec 2022 04:26:04 PM CST
Vds_pch18_gm_gds.csv	135.1 KB	CSV document	Thu 29 Dec 2022 04:26:35 PM CST
Vds_pch18_gm_Id.csv	134.3 KB	CSV document	Thu 29 Dec 2022 04:27:00 PM CST

2、运行 nch18_GmId_readfiles.m，并检查.mat 文件中的数据是否已经更新

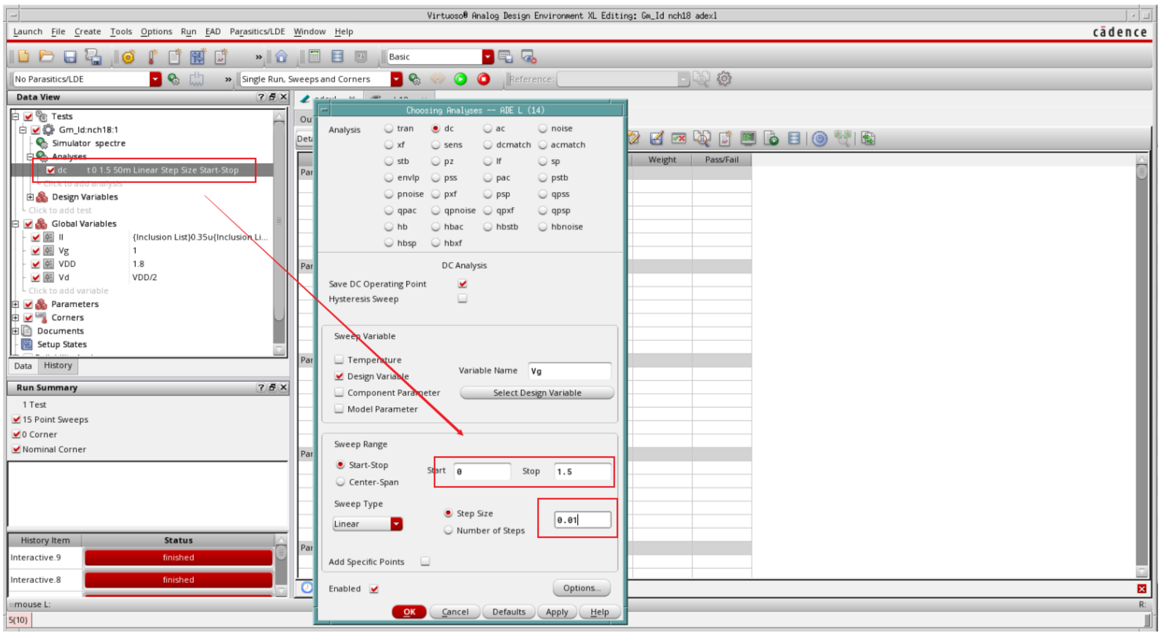
3、运行 MOS_GmId_GUI.m 并加载数据即可正常使用

数据扫描过程

1、以 mos18 为例，新建 Sch 文件，一共三个变量：ll、Vg、Vd



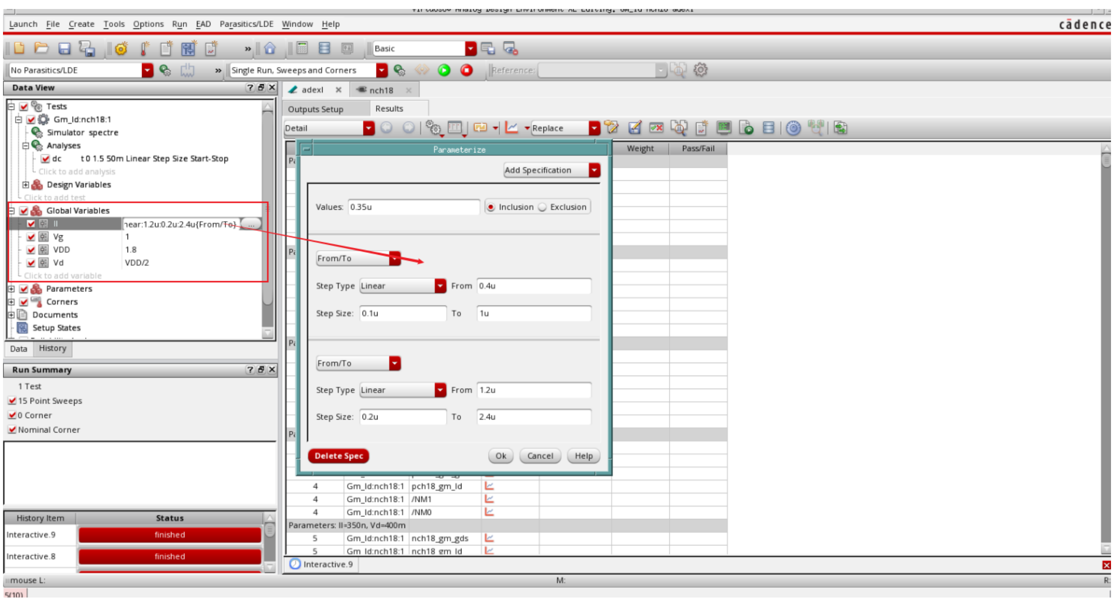
2、新建 ADE XL 文件，设置 dc 直流仿真的变量扫描



3、跑一个直流仿真之后，通过 calculator->OS 添加 8*2 个输出

Test	Name	Type	Details	EvalType	Plot	Save
Gm_Id:nch...	nch18_Cdd_Cgg	expr	(OS("/NM0" "cdd") / OS("/NM0" "cgg"))	point	☐	☐
Gm_Id:nch...	nch18_Cgd_Cgg	expr	(OS("/NM0" "cgd") / OS("/NM0" "cgg"))	point	☐	☐
Gm_Id:nch...	nch18_ft	expr	(OS("/NM0" "gm") / OS("/NM0" "cgs") / 2 / 3.14)	point	☐	☐
Gm_Id:nch...	nch18_gm_gds	expr	OS("/NM0" "self_gain")	point	☑	☐
Gm_Id:nch...	nch18_Id_W	expr	(OS("/NM0" "id") / 1e-06)	point	☐	☐
Gm_Id:nch...	nch18_Vdsat	expr	OS("/NM0" "vdsat")	point	☐	☐
Gm_Id:nch...	nch18_Vgs_Vth	expr	(OS("/NM0" "vgs") - OS("/NM0" "vth"))	point	☐	☐
Gm_Id:nch...	nch18_gm_Id	expr	OS("/NM0" "gmoverid")	point	☑	☐
Gm_Id:nch...	pch18_Cdd_Cgg	expr	(OS("/NM1" "cdd") / OS("/NM1" "cgg"))	point	☐	☐
Gm_Id:nch...	pch18_Cgd_Cgg	expr	(OS("/NM1" "cgd") / OS("/NM1" "cgg"))	point	☐	☐
Gm_Id:nch...	pch18_ft	expr	(OS("/NM1" "gm") / OS("/NM1" "cgs") / 2 / 3.14)	point	☐	☐
Gm_Id:nch...	pch18_gm_gds	expr	OS("/NM1" "self_gain")	point	☑	☐
Gm_Id:nch...	pch18_Id_W	expr	(OS("/NM1" "id") / 1e-06)	point	☐	☐
Gm_Id:nch...	pch18_Vdsat	expr	OS("/NM1" "vdsat")	point	☐	☐
Gm_Id:nch...	pch18_gm_Id	expr	OS("/NM1" "gmoverid")	point	☑	☐
Gm_Id:nch...	pch18_Vgs_Vth	expr	(OS("/NM1" "vgs") - OS("/NM1" "vth"))	point	☐	☐

4、设置变量 Il 的嵌套扫描范围和步长



5、运放仿真得到 16 组扫描数据，依次保存对应的.csv 文件名

nch18_Cdd_Cgg.csv	81.3 KB	CSV document	Thu 29 Dec 2022 09:41:11 AM CST
nch18_Cgd_Cgg.csv	83.5 KB	CSV document	Thu 29 Dec 2022 01:47:46 PM CST
nch18_ft.csv	79.4 KB	CSV document	Thu 29 Dec 2022 01:50:53 PM CST
nch18_gm_gds.csv	77.3 KB	CSV document	Thu 29 Dec 2022 02:31:41 PM CST
nch18_gm_Id.csv	77.3 KB	CSV document	Thu 29 Dec 2022 10:46:04 AM CST
nch18_Id_W.csv	79.6 KB	CSV document	Thu 29 Dec 2022 01:52:12 PM CST
nch18_Vdsat.csv	80.3 KB	CSV document	Thu 29 Dec 2022 01:57:29 PM CST
nch18_Vgs_Vth.csv	80.6 KB	CSV document	Thu 29 Dec 2022 01:59:54 PM CST
pch18_Cdd_Cgg.csv	81.1 KB	CSV document	Thu 29 Dec 2022 03:05:22 PM CST
pch18_Cgd_Cgg.csv	83.4 KB	CSV document	Thu 29 Dec 2022 03:06:14 PM CST
pch18_ft.csv	79.4 KB	CSV document	Thu 29 Dec 2022 03:07:38 PM CST
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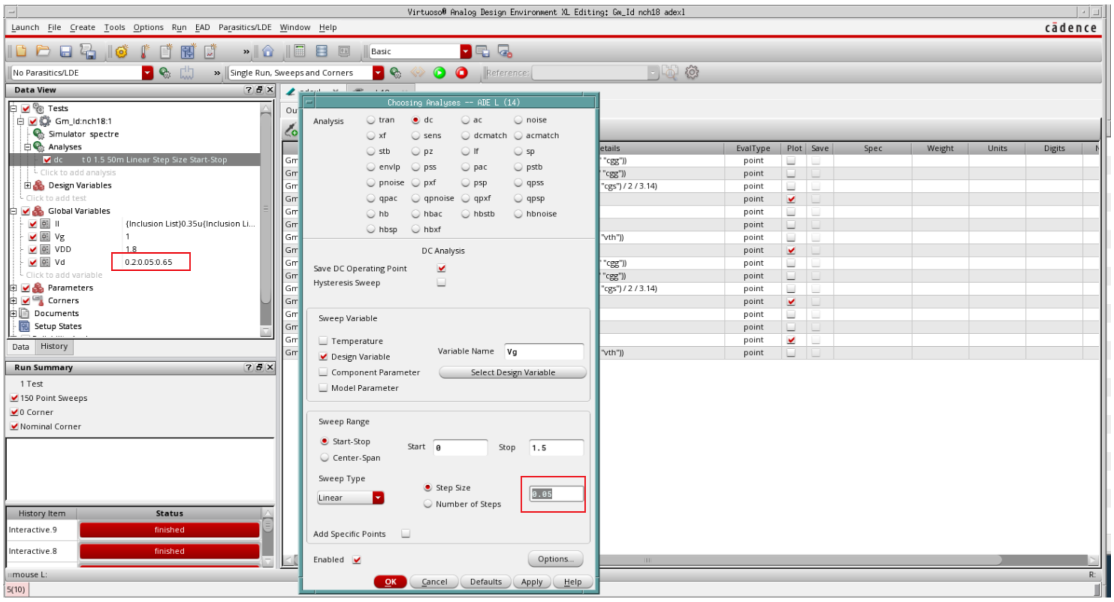
pch18_gm_Id.csv	77.2 KB	CSV document	Thu 29 Dec 2022 03:09:36 PM CST
pch18_Id_W.csv	82.2 KB	CSV document	Thu 29 Dec 2022 03:08:37 PM CST
pch18_Vdsat.csv	82.5 KB	CSV document	Thu 29 Dec 2022 03:09:01 PM CST
pch18_Vgs_Vth.csv	81.3 KB	CSV document	Thu 29 Dec 2022 03:09:58 PM CST

6、将 dc 变量扫描步长改为 0.05，Vd 添加二层嵌套扫描 0.2 : 0.05 : 0.65，输出只需要勾选 gm_Id 和 gm_gds,得到四组 Vds 数据，保存对应的.cds 文件名



Vds_nch18_gm_gds.csv	135.1 KB	CSV document	Thu 29 Dec 2022 04:25:26 PM CST
Vds_nch18_gm_Id.csv	134.4 KB	CSV document	Thu 29 Dec 2022 04:26:04 PM CST
Vds_pch18_gm_gds.csv	135.1 KB	CSV document	Thu 29 Dec 2022 04:26:35 PM CST
Vds_pch18_gm_Id.csv	134.3 KB	CSV document	Thu 29 Dec 2022 04:27:00 PM CST